

4. Grid Convergence

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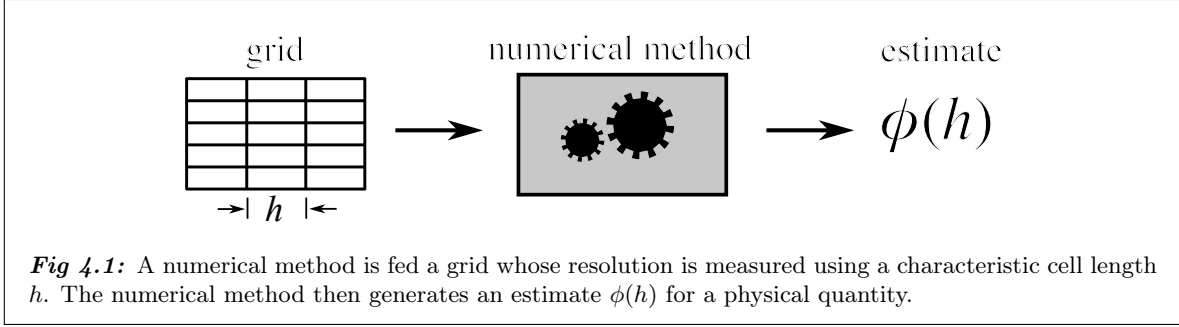
Introduction

CFD simulations rely on the assumption that as a grid is refined, the numerical approximation converges to an exact solution of the mathematical model. In Section 3.5, we used this assumption to verify a code against a manufactured analytical solution. In practice, however, CFD is typically used to simulate problems for which there is no known analytical solution, and for which we cannot verify our simulations directly. We can, however, still check whether our numerical predictions tend to a hypothetical “grid-independent” result as the grid is refined. Over the years, the CFD community has established a set of best practices, called a *grid convergence study*, for checking this convergence. The procedure measures the change in simulated results across a sequence of three or more progressively refined grids. Using that data, it estimates the hypothetical grid-independent result in the limit of infinite resolution, and characterizes how quickly simulated results approach that limit with an *observed order-of-accuracy*. Finally, it estimates the uncertainty of the finite-grid results by computing a *Grid Convergence Index* (GCI).

Grid convergence studies are an important issue in CFD, because simulated fluid flows can change drastically as a grid is refined. Most CFD analysts have experienced a simulation that produces reasonable results on a coarse grid, only for the flow to unexpectedly transition from steady to unsteady flow, or even laminar to turbulent flow, as the grid is refined.

Source Note and Further Reading

Grid convergence studies are widely discussed in the literature, but treatments vary and are not always consistent with one another. This chapter offers an approachable introduction to the topic, primarily following the presentation of Celik *et al.* [7], supplemented by references [8, 9, 10].



4.1 Preliminary assumptions and definitions

Suppose our objective is to approximate a physical quantity with a single scalar value that does not vary in time or space. For example, we might be interested in approximating the maximum temperature T_{max} in a steady-state heat conduction problem, as opposed to a spatially varying temperature field $T(x, y)$. We estimate the quantity using the procedure sketched in Fig. 4.1, in which a numerical method is “fed” a grid whose resolution is measured using a characteristic cell size h (defined as a length). The method then outputs an estimate $\phi(h)$ of the physical quantity. Using this thought problem, a grid convergence study makes two assumptions, labeled (i) and (ii) below.

- (i) In the hypothetical limit of an infinitely fine grid ($h \rightarrow 0$), the estimate $\phi(h)$ converges to a limiting value ϕ_o ,

$$\lim_{h \rightarrow 0} \phi(h) = \phi_o. \quad (4.1)$$

Definition 4.1: Discretization Error

We define the *discretization error* $e(h)$ as the difference between the estimate $\phi(h)$ on a finite grid, and the hypothetical limit ϕ_o ,

$$e(h) = \phi(h) - \phi_o. \quad (4.2)$$

Note that $e(h)$ only measures errors due to grid resolution. Furthermore, ϕ_o may not equal the exact value of the estimated quantity, due to other sources of error.

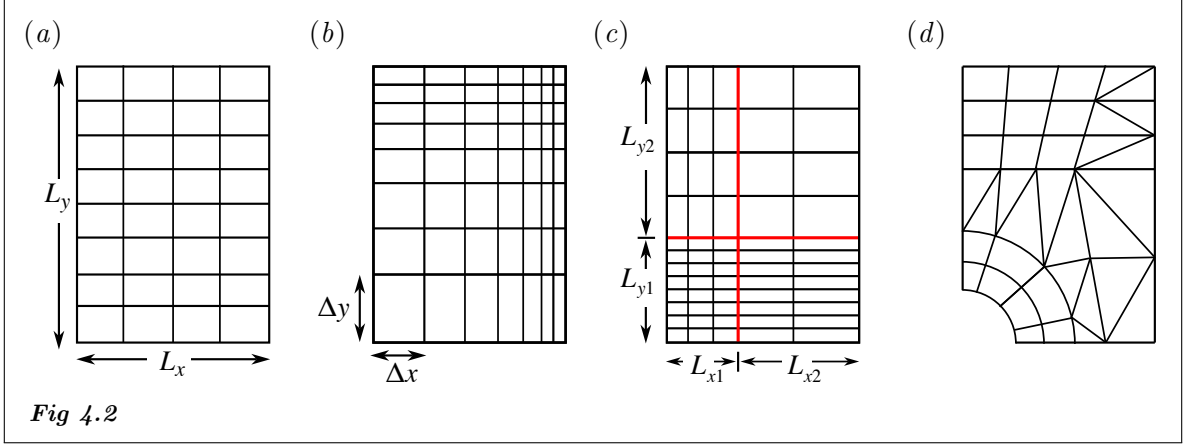
- (ii) On sufficiently fine grids, the estimate $\phi(h)$ approaches ϕ_o asymptotically as

$$\phi(h) = \phi_o + Ch^p, \quad \text{or equivalently, } e(h) = Ch^p, \quad (4.3)$$

where C is a constant.

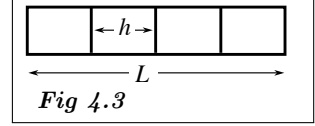
Definition 4.2: Observed Order-of-Accuracy

The exponent p in Eq. (4.3) is called the *observed order-of-accuracy*, and may differ from the expected order p^* of the underlying numerical method.



4.1.1 Choosing a characteristic cell size h

We now turn to the practical question of how to define the characteristic cell size h . We begin with the simplest case of a uniform one-dimensional grid, as sketched in Fig. 4.3. In this case, an obvious choice for h is

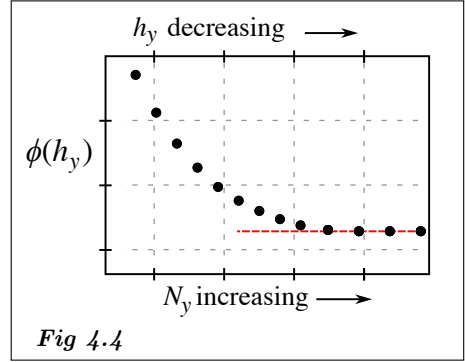


$$h = \frac{L}{N}, \quad (4.4)$$

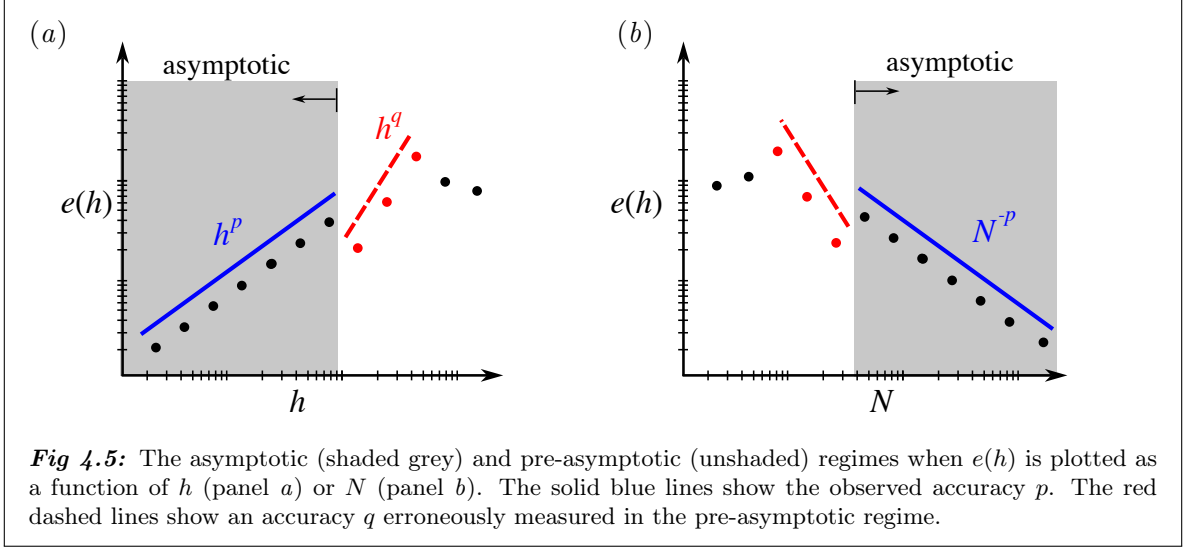
where L is the domain length and N is the total number of cells. Next, we consider the 2D uniform grid sketched in Fig. 4.2(a), which discretizes the domain $L_x \times L_y$ with $N_x = 4$ cells in the x -direction and $N_y = 8$ cells in the y -direction. Due to the added dimension, two potential definitions of h now include

$$h_x = \frac{L_x}{N_x} \quad \text{and} \quad h_y = \frac{L_y}{N_y}. \quad (4.5)$$

Typically, either choice is acceptable, provided the ratio N_x/N_y is fixed as the grid is refined, which ensures both h_x and h_y tend to zero. To see why this is necessary, suppose we set $h = h_y$, and then refine the grid by increasing N_y while holding N_x fixed. The resulting estimate $\phi(h_y)$ may approach a constant value as $N_y \rightarrow \infty$, as sketched in Fig. 4.4. However, this limit is not ϕ_o . Rather, it is the value ϕ approaches when N_x is held fixed. Moreover, the apparent convergence with N_y is likely masking a poor resolution in the x -direction.



Now consider the non-uniform grid in Fig. 4.2(b), where the cell size Δx decreases with increasing x and the cell size Δy decreases with increasing y . In practice, if the non-uniformity is gradual, it is usually acceptable to set h to either L_x/N_x or L_y/N_y , as for a uniform grid, assuming again that the ratio N_x/N_y is fixed. Fig. 4.2(c) shows a second type of non-uniform grid, in which the domain is partitioned into four sub-domains with lengths L_{x1} and L_{x2} in the x -direction and L_{y1} and L_{y2} in the y -direction. Each sub-domain is discretized with its own uniform resolution N_{x1} , N_{x2} , N_{y1} , and N_{y2} , respectively. In this case, provided the ratios between sub-region resolutions are held fixed during refinement, we can again define h using any one sub-region, for example $h = L_{x1}/N_{x1}$.



For the Cartesian grids considered in this textbook, a characteristic cell size can be defined using the approaches above. The definition of h is more challenging, however, for general unstructured grids like that sketched in Fig. 4.2(d). Here we only note that Celik *et al.* [7] suggest the following measures h_{2D} and h_{3D} for 2D and 3D unstructured grids, respectively,

$$h_{2D} = \left[\frac{1}{N} \sum_{i=1}^N A_i \right]^{1/2}, \quad h_{3D} = \left[\frac{1}{N} \sum_{i=1}^N V_i \right]^{1/3}, \quad (4.6)$$

where A_i and V_i are the area and volume, respectively, of the i 'th cell in a grid with N cells.

4.1.2 The asymptotic regime

In practice, the power-law relationship $e(h) = Ch^p$ only holds when h is sufficiently small, say $h < h^*$. We refer to this range of grid resolutions as the **asymptotic regime**. Figure 4.5(a) illustrates this behavior by plotting $e(h)$ as a function of h , while panel (b) shows the corresponding behavior when $e(h)$ is plotted as a function of N , assuming $h \sim 1/N$. In the asymptotic regime (shaded grey), the error decreases smoothly with grid refinement following the order-of-accuracy p (blue line). However, on coarse grids in the pre-asymptotic regime $h > h^*$, the error can vary unpredictably, and even increase with refinement. One dangerous situation to keep in mind is that computed values of $e(h)$ in the pre-asymptotic regime can appear well-behaved. For example, the red data points in Fig. 4.5 appear to follow the power law shown as a red dashed line. If a practitioner ran only these three grid resolutions, they might erroneously conclude their method is converging at the order q .

For the cases encountered in this textbook, the pre-asymptotic regime is typically small, and only modest resolutions are required to reach the asymptotic regime. In more general CFD practice, however, the minimum resolution h^* may exceed a user's computing resources. Those challenging cases force a choice between acquiring more computing resources or simplifying the problem.

Side Note 4.1: The asymptotic regime in turbulent flow simulations.

A striking example of the asymptotic regime arises in the direct numerical simulation (DNS) of turbulent flows. In a turbulent flow, energy is transferred from large eddies down to progressively smaller eddies, until it is dissipated as heat at the smallest scales. The size of these smallest eddies decreases with increasing Reynolds number, so that accurately resolving all scales of motion requires a grid that becomes finer and finer as the Reynolds number increases. For high Reynolds number flows, the required cell count can easily exceed the capacity of even the largest available supercomputers. For this reason, DNS is currently limited to sufficiently low Reynolds numbers and simple geometries. For more complex geometries and high Reynolds number flows, alternative modeling approaches must be used, such as Reynolds-Averaged Navier–Stokes (RANS) equations, or Large Eddy Simulation (LES).

4.2 Estimating the observed order of accuracy

In the asymptotic regime, we can estimate the observed order-of-accuracy p by computing ϕ on three progressively finer grids. Consider three grids with the cell sizes $h_1 < h_2 < h_3$, and corresponding solutions ϕ_1 , ϕ_2 , and ϕ_3 . Assuming the grids are in the asymptotic regime, we can solve the three equations below,

$$\phi_1 = \phi_o + Ch_1^p, \quad (4.7)$$

$$\phi_2 = \phi_o + Ch_2^p, \quad (4.8)$$

$$\phi_3 = \phi_o + Ch_3^p, \quad (4.9)$$

for the three unknowns: ϕ_o , C , and p . For that, Celik *et al.* [7] first define the differences e_{21} and e_{32} below,

$$e_{21} = \phi_2 - \phi_1 = C(h_2^p - h_1^p), \quad (4.10)$$

$$e_{32} = \phi_3 - \phi_2 = C(h_3^p - h_2^p), \quad (4.11)$$

and take the ratio e_{32}/e_{21} to eliminate the constant coefficient C ,

$$\frac{e_{32}}{e_{21}} = \frac{h_3^p - h_2^p}{h_2^p - h_1^p}. \quad (4.12)$$

With a little algebra, they express the result as

$$\frac{e_{32}}{e_{21}} = \frac{r_{21}^p (r_{32}^p - 1)}{r_{21}^p - 1}, \quad (4.13)$$

where they define the refinement ratios

$$r_{21} = \frac{h_2}{h_1}, \quad r_{32} = \frac{h_3}{h_2}. \quad (4.14)$$

For the scope of this textbook, we assume $r_{21} = r_{32} = r$.¹ In that case, we can solve Eq. (4.13) for p as,

$$p = \frac{1}{\ln(r)} \ln \left(\frac{e_{32}}{e_{21}} \right). \quad (4.15)$$

¹The general case where $r_{21} \neq r_{32}$ requires special treatment, as discussed in Celik *et al.* [7].

At this point, it's important to note that Eq. (4.15) is undefined when the ratio e_{32}/e_{21} is negative. The reason is that if $e(h)$ varies as Ch^p , the differences e_{21} and e_{32} defined in Eqs. (4.10)–(4.11) are either both positive when $C > 0$ or both negative when $C < 0$. In practice, however, this assumption may not hold, and negative ratios e_{32}/e_{21} can occur. Celik *et al.* [7] deal with those cases by taking the absolute value of e_{32}/e_{21} when computing p , as below

$$p = \frac{1}{\ln(r)} \ln \left| \frac{e_{32}}{e_{21}} \right|. \quad (4.16)$$

Equation (4.16) is the form found in most sources. We discuss this issue further in section 4.3 below.

4.3 Convergence types

When interpreting your results for the observed order p , it can be useful to consider the “convergence ratio” R , defined as

$$R = \frac{e_{21}}{e_{32}} = \frac{\phi_2 - \phi_1}{\phi_3 - \phi_2}. \quad (4.17)$$

Using this ratio, one can define the four different types of convergence behavior [8, 9] illustrated in Fig. 4.6 and discussed below.

Monotonic convergence: $0 < R < 1$

Monotonic convergence refers to cases for which $\phi(h)$ approaches ϕ_o monotonically from above or below, as sketched in Fig. 4.6(a). Monotonic convergence is the best-case scenario, because it is consistent with the assumption that $e(h)$ varies asymptotically as

$$e(h) = \phi(h) - \phi_o = Ch^p. \quad (4.18)$$

This power law approaches ϕ_o monotonically “from above” when $C > 0$, and “from below” when $C < 0$.

Oscillatory convergence: $-1 < R < 0$

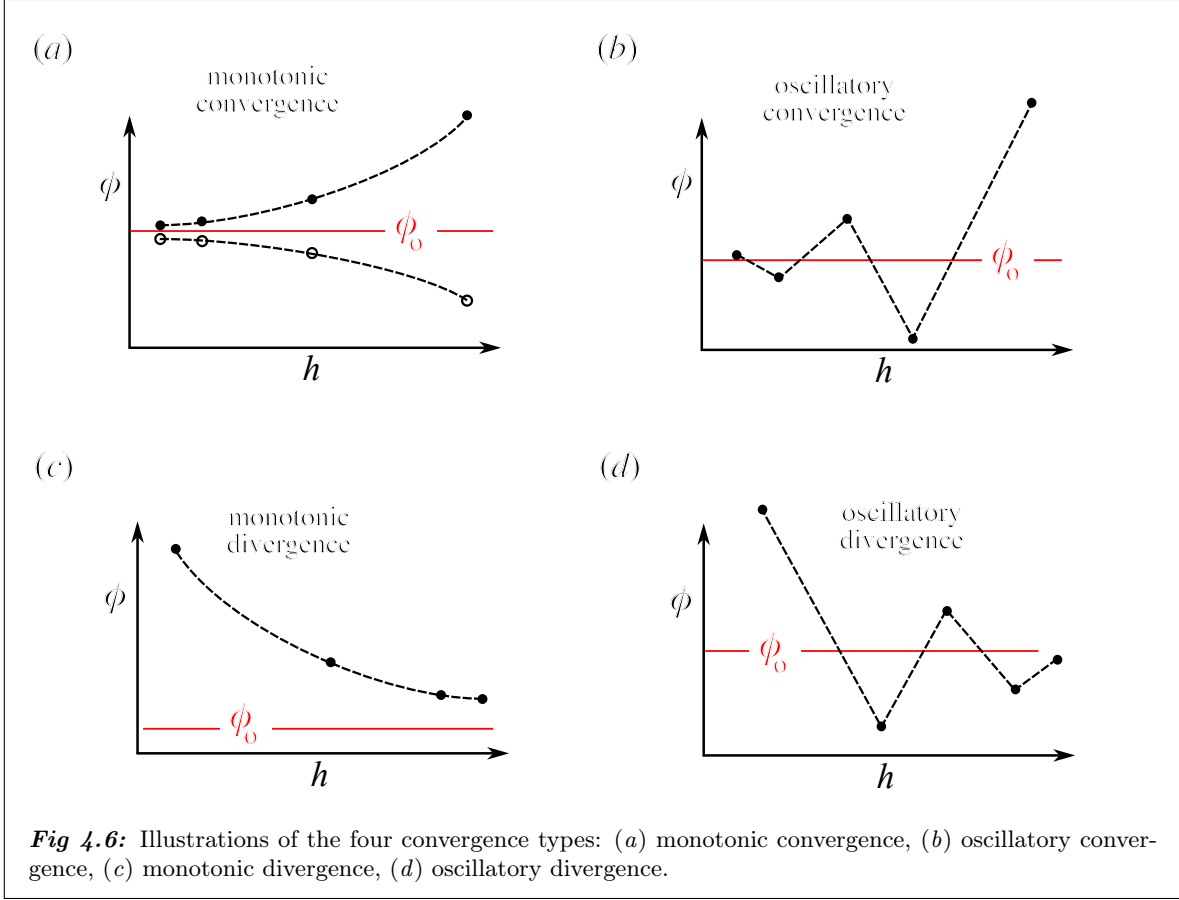
Oscillatory convergence refers to cases in which $\phi(h)$ approaches ϕ_o in an oscillatory manner, as in Fig. 4.6(b). This causes a sign change between e_{32} and e_{21} , and violates our assumption that $e(h)$ varies as

$$e(h) = Ch^p. \quad (4.19)$$

In some cases, oscillatory convergence indicates that one or more of the resolutions (h_1 , h_2 , h_3) lies outside the asymptotic regime. To investigate that possibility, you can shift and refine the three resolutions by dropping the data at h_3 , setting the old h_2 and h_1 to the new h_3 and h_2 , respectively, and running a new simulation at a finer h_1 . If oscillatory behavior persists, but the data suggests the *absolute* error $|e(h)|$ decreases asymptotically,

$$|e(h)| = Ch^p, \quad (4.20)$$

it is common practice to accept the p value computed using Eq. (4.16). In that case, however, the discussion of your results should mention that the convergence was oscillatory.



Monotonic and oscillatory divergence: $|R| > 1$

In some cases, the absolute value $|R|$ is greater than one, as in Figs. 4.6(c) and 4.6(d). In the best case scenario, this indicates that one or more of your resolutions is outside the asymptotic regime, which you can investigate by shifting and refining the resolutions. However, if the behavior persists, there may be a fundamental issue in the mathematical model or numerical method that needs to be resolved. Note that some sources classify the divergence as *monotonic divergence* when $R > 1$ (Fig. 4.6c) and *oscillatory divergence* when $R < -1$ (Fig. 4.6d).

4.4 Consistency check

Even for cases of monotonic convergence, it is important to consider whether the three grid resolutions (h_1, h_2, h_3) are truly in the asymptotic regime. An initial check is to compare the observed order-of-accuracy p with the expected order p^* of the numerical method. For example, if the expected order is second-order ($p^* = 2$), a measured accuracy of $p = 1.95$ adds confidence that the three grids were likely in the asymptotic regime. On the other hand, unexpected p -values, such as $p = 1.1$ or $p = 3.5$, suggest you should recompute p after shifting and refining the grid.

If, after shifting and refining the grid, p converges to a stable value that differs significantly from p^* , then it is important to consider that there are valid reasons why the observed order p may differ from the expected p^* , even in the asymptotic regime. For example, the expected

p^* may rely on Taylor series approximations that assume a smoothly varying, infinitely differentiable, flow field, while the actual problem includes a discontinuous flow field or boundary condition. These issues can easily reduce an expected second-order method to first-order.²

4.5 Estimating discretization error and uncertainty

After computing the observed order p , Celik *et al.* [7] recommend estimating and reporting the relative discretization error using two measures. The first is the *approximate relative error* e_a^{21} defined below,

$$e_a^{21} = \frac{|\phi_2 - \phi_1|}{|\phi_1|} \quad \text{approximate relative error.} \quad (4.21)$$

Though simple to compute, e_a^{21} is an unreliable measure of discretization error. First, it measures the difference $|\phi_1 - \phi_2|$, rather than $|\phi_1 - \phi_o|$. Second, it can be made arbitrarily small by setting the refinement ratio r close to one.

For a more reliable measure of discretization error, we can measure the error relative to the ϕ_o value computed by solving equations (4.7)–(4.9). Celik *et al.* denote this value

$$\phi_{ext}^{21} = \frac{r^p \phi_1 - \phi_2}{r^p - 1}, \quad (4.22)$$

where the subscript “*ext*” implies this value is extrapolated using ϕ_1 and ϕ_2 . This nomenclature, however, can cause some confusion, as detailed in the side note below. Using this extrapolated estimate to ϕ_o , Celik *et al.* recommend computing the *extrapolated relative error* defined below,

$$e_{ext}^{21} = \frac{|\phi_{ext}^{21} - \phi_1|}{|\phi_{ext}^{21}|} \quad \text{extrapolated relative error.} \quad (4.23)$$

Side Note 4.2: An issue regarding ϕ_{ext}^{21} .

In addition to ϕ_{ext}^{21} , Celik *et al.* [7] recommend computing the analogous quantity

$$\phi_{ext}^{32} = \frac{r^p \phi_2 - \phi_3}{r^p - 1}, \quad (4.24)$$

extrapolated using ϕ_2 and ϕ_3 . However, assuming monotonic convergence, these two values are exactly equal, $\phi_{ext}^{21} = \phi_{ext}^{32}$, when p is the observed order computed from Eqs. (4.7)–(4.9). In that case, ϕ_{ext}^{21} and ϕ_{ext}^{32} are two algebraically equivalent expressions for the same ϕ_o obtained by solving equations (4.7)–(4.9). On the other hand, the two values may differ if p is obtained independently, such as by setting p to the expected order p^* .

In addition to e_a^{21} and e_{ext}^{21} , Celik *et al.* [7] recommend reporting the overall “*uncertainty*” due to discretization via a *grid convergence index* (GCI). The GCI differs from the error estimates e_a^{21} and e_{ext}^{21} in the sense that it estimates a band about ϕ_1 in which ϕ_o is expected

²That does not mean your simulated results are necessarily wrong, and depending on the case, you might choose to live with the reduced accuracy by simply using finer grids than initially hoped.

to lie. To motivate its definition, we first observe that Eq. (4.22) can be expressed in the form below,

$$\phi_{ext}^{21} = \phi_1 + \underbrace{\frac{\phi_1 - \phi_2}{r^p - 1}}_{\text{correction}}, \quad (4.25)$$

where the second term on the right-hand-side is interpreted as a correction to ϕ_1 . The magnitude of this correction relative to ϕ_1 can be measured as

$$\frac{|\phi_{ext}^{21} - \phi_1|}{|\phi_1|} = \frac{1}{|\phi_1|} \frac{|\phi_1 - \phi_2|}{r^p - 1} = \frac{e_a^{21}}{r^p - 1}. \quad (4.26)$$

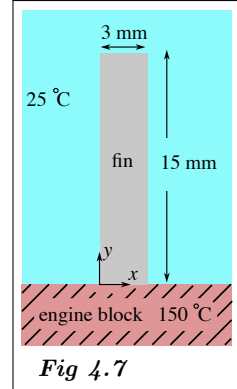
Celik *et al.* define the GCI as this measure with an added safety factor $F_s = 1.25$,

$$\boxed{\text{GCI}^{21} = F_s \frac{e_a^{21}}{r^p - 1}}. \quad (4.27)$$

4.6 Case study: Heat fin

CFD engineers are often asked to simulate a component or process within a larger system. When the surrounding system is too large to simulate directly, the engineer must truncate the domain and approximate the surroundings through boundary conditions. This modeling step often “quietly” degrades the accuracy of a simulated quantity $\phi(h)$, in the sense that the simulated flow fields appear physically reasonable, giving no obvious sign that something is wrong.

Grid convergence studies are particularly useful for rooting out a subtle loss of accuracy that isn’t otherwise obvious from the flow fields. For demonstration, here we consider the simulation of the steady temperature distribution in an aluminum fin that dissipates heat from an engine block to surrounding air, as sketched in Fig. 4.7. Rather than simulating the surrounding air directly, we truncate the computational domain to the fin itself, defined by $x \in [0, L_x]$ and $y \in [0, L_y]$, where $L_x = 3$ mm and $L_y = 15$ mm. We assume the temperature field in the fin is governed by the 2D diffusion equation in the $x - y$ plane,



$$\int_{CS} k(\nabla T \cdot \mathbf{n}) dA = 0, \quad (4.28)$$

where $k = 200$ W/m/K is the thermal conductivity of aluminum.

When performing a grid convergence study, it’s important to set $\phi(h)$ to a quantity that reflects the actual goal of the simulation [7]. For our discussion, we assume our engineering objective is to predict the rate of heat transfer (per unit depth in the z -direction) from the engine block into the fin,

$$\dot{Q} = \int_0^{L_x} \left(-k \frac{\partial T}{\partial y} \Big|_{y=0} \right) dx. \quad (4.29)$$

To demonstrate how inconsistent boundary conditions can degrade accuracy, suppose we model the adjacent engine block by applying a constant block temperature $T_b = 150$ °C along the southern boundary ($y = 0$) of the fin,

$$T \Big|_{y=0} = T_b, \quad (4.30)$$

N	$\dot{Q}$ (kW/m)	p	e_a^{21} (%)	ϕ_{ext}^{21} (kW/m)	e_{ext}^{21} (%)	GCI^{21} (%)
4	36.0	—	—	—	—	—
8	59.5	—	—	—	—	—
16	83.8	-0.0402	28.9	-7.98×10^2	111	-1.32×10^3
32	106	0.0976	21.3	4.30×10^2	75.2	3.80×10^2
64	129	0.0294	17.3	1.20×10^3	89.3	1.05×10^3
128	151	0.00619	14.7	5.29×10^3	97.1	4.26×10^3
256	173	0.00150	12.8	2.14×10^4	99.2	1.54×10^4
512	195	0.000241	11.3	1.32×10^5	99.9	8.48×10^4
1024	217	0.000272	10.2	1.17×10^5	99.8	6.74×10^4

Table 4.1: Grid convergence study for $\dot{Q}$ using the fixed-temperature boundary conditions (4.31).

and we model the surrounding air by applying a constant air temperature $T_{air} = 25$ °C on the remaining three boundaries of the fin,

$$T\Big|_{x=0} = T\Big|_{x=L_x} = T\Big|_{y=L_y} = T_{air}. \quad (4.31)$$

These conditions are inconsistent at the two corners where the southern boundary meets the east/west boundaries.

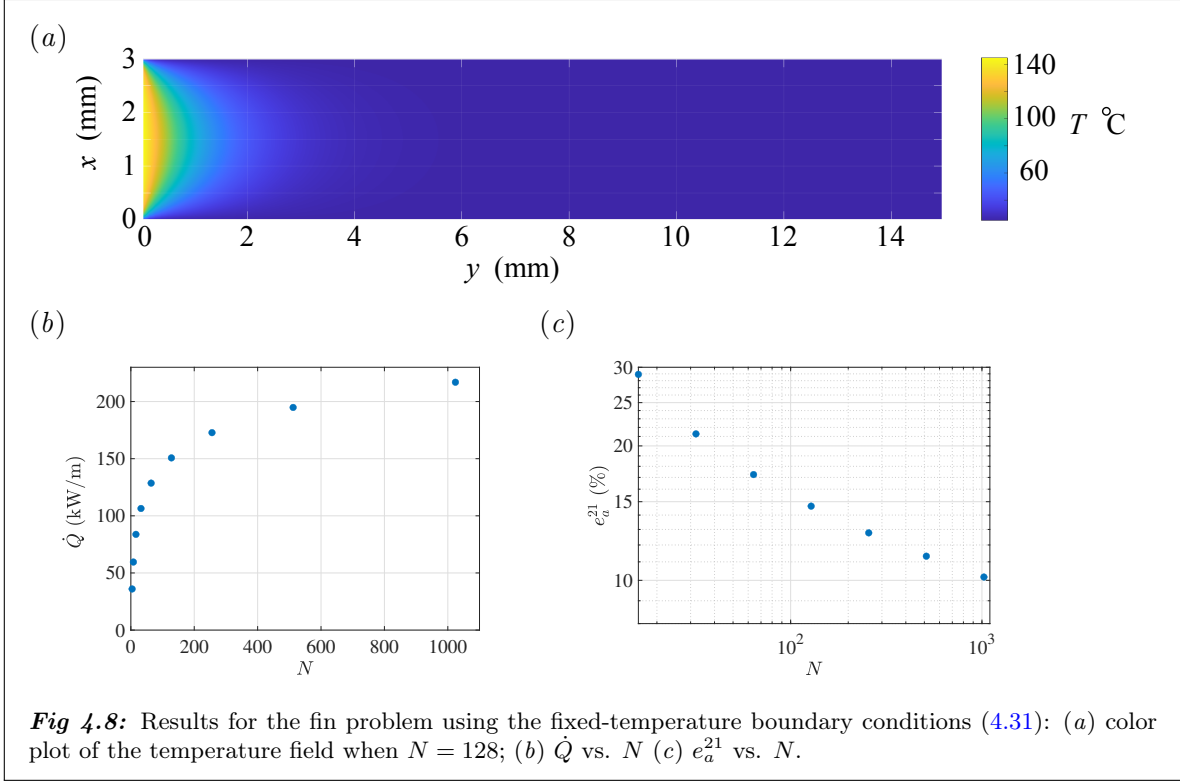
The numerical simulation of this problem is the topic of a homework problem. To perform a grid convergence study, we simulate the fin using a uniform grid with $N_x = N_y = N$, for the nine grid resolutions listed in the first column of Table 4.1, ranging from $N = 4$ to $N = 1024$. Each successive resolution doubles N , giving a refinement ratio $r = 2$. Using the simulated $\dot{Q}$ values in the second column, we compute the observed order of accuracy p from Eq. (4.16) for each set of three successive grids ($N = 4, 8, 16$, then $N = 8, 16, 32$, and so on up to $N = 256, 512, 1024$), and list each value of p next to the finest resolution used. The remaining columns list e_a^{21} , ϕ_{ext}^{21} , e_{ext}^{21} , and GCI^{21} , each computed from the same triple of grids as p .

Figure 4.8(a) shows a color plot of the simulated temperature field at $N = 128$. We have rotated the plot 90° clockwise to accommodate the large aspect ratio L_y/L_x . The temperature field appears smooth and physically reasonable, with no visual hint that anything is wrong.³ Panel (b), however, shows that $\dot{Q}$ remains visually sensitive to grid resolution out to $N = 1024$, which is an absurdly fine grid for such a simple geometry. Panel (c) also shows that though the approximate relative error e_a^{21} decreases monotonically with N , it only reaches 10.2% by $N = 1024$. If we judge convergence using e_a^{21} , and a 10% error is acceptable, we might be tempted to accept the result at $N = 1024$. If, however, we consider the remaining results in Table 4.1, we see that the observed order p tends to zero with increasing resolution, and the extrapolated estimate ϕ_{ext}^{21} varies over three orders of magnitude, from about 10^2 to 10^5 kW/m. Meanwhile e_{ext}^{21} remains near 100%, and GCI^{21} tends to increase with N . Something is clearly wrong, and we should not trust our simulated $\dot{Q}$, even at $N = 1024$, as a rough estimate.

Rather than applying T_{air} directly on the fin surface, as in Eq. (4.31), CFD modelers more commonly approximate the local heat transfer from a fin surface to adjacent air using a convective condition of the form

$$-k \mathbf{n} \cdot \nabla T\Big|_{surf} = h(T_{surf} - T_{air}), \quad (4.32)$$

³Here we do not assume you already know that when $L_y \gg L_x$, the temperature field should vary primarily in the axial y -direction and should be nearly uniform over any transverse cross section.



where the subscript *surf* denotes the fin surface in contact with air, h is a convective heat transfer coefficient, and T_{air} is the air temperature far from the fin. Because T_{surf} is unknown, we apply condition (4.32) as an equivalent Robin condition,

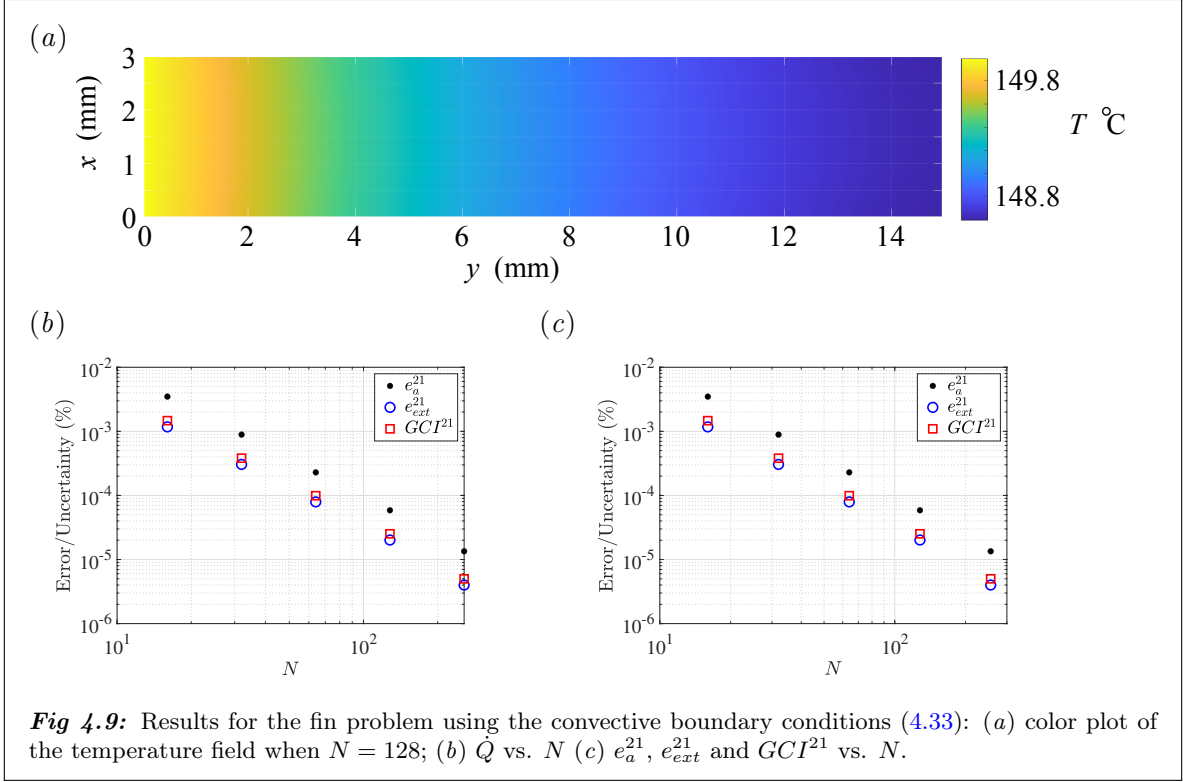
$$\left[-k \mathbf{n} \cdot \nabla T - hT \right]_{surf} = -hT_{air}. \quad (4.33)$$

By not setting a fixed temperature on the fin surface, this condition resolves the inconsistency at the two corners where air, fin, and block meet.

Table 4.2 shows the results of a grid convergence study using the convective boundary conditions (4.33) with $h = 25 \text{ W/m}^2/\text{K}$ and the resolutions $N = 4$ through $N = 256$. The simulated heat transfer rate $\dot{Q}$ in the second column now varies only after the fourth significant digit across the full range of resolutions. Figure 4.9(b) shows $\dot{Q}$ vs. N , confirming that $\dot{Q}$

N	$\dot{Q}$ (W/m)	p	e_a^{21} (%)	ϕ_{ext}^{21} (W/m)	e_{ext}^{21} (%)	GCI^{21} (%)
4	102.328108	—	—	—	—	—
8	102.342267	—	—	—	—	—
16	102.345837	1.99	3.49×10^{-3}	102.347039	1.18×10^{-3}	1.47×10^{-3}
32	102.346748	1.97	8.90×10^{-4}	102.347060	3.05×10^{-4}	3.81×10^{-4}
64	102.346982	1.96	2.29×10^{-4}	102.347063	7.93×10^{-5}	9.92×10^{-5}
128	102.347042	1.97	5.86×10^{-5}	102.347063	2.02×10^{-5}	2.52×10^{-5}
256	102.347056	2.12	1.34×10^{-5}	102.347060	4.00×10^{-6}	5.00×10^{-6}

Table 4.2: Grid convergence study for $\dot{Q}$ in the fin problem with the convective boundary condition (4.33).



quickly plateaus with grid refinement. Figure 4.9(a) shows the simulated temperature field at $N = 128$. The temperature field now varies negligibly in the transverse x -direction, and is essentially one-dimensional in y . Figure 4.9(c) shows e_a^{21} (solid black dots), e_{ext}^{21} (blue circles), and GCI^{21} (red squares) vs. N . All three quantities decrease monotonically with increasing resolution, and are already of order $10^{-3}\%$ by $N = 16$.